

# A Root-Complete Cnoidal and Soliton Atlas for KdV Traveling Waves

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## Abstract

We classify every bounded classical real traveling profile of the plus-sign Korteweg–de Vries equation directly from its cubic first integral. Three simple real roots give one translated Jacobi-cn<sup>2</sup> family with exact speed and fundamental period; a lower double root gives the sech<sup>2</sup> soliton, and all remaining root topologies give only bounded constants.

## 1 Frozen equation and cubic reduction

We fix

$$u_t + 6uu_x + u_{xxx} = 0, \quad u(x, t) = U(\xi), \quad \xi = x - ct, \quad (1)$$

with  $U \in C^3(\mathbb{R}; \mathbb{R})$  bounded. Integrating the traveling equation twice gives

$$U'' = cU - 3U^2 + A, \quad (U')^2 = -2U^3 + cU^2 + 2AU + B. \quad (2)$$

If the cubic has real roots  $r_1 \leq r_2 \leq r_3$ , its compact allowed branch has the convention

$$(U')^2 = 2(r_3 - U)(U - r_2)(U - r_1), \quad (3)$$

and coefficient comparison forces

$$c = 2(r_1 + r_2 + r_3), \quad A = -(r_1r_2 + r_1r_3 + r_2r_3), \quad B = 2r_1r_2r_3. \quad (4)$$

These signs are part of the frozen object: switching the sign of  $6uu_x$  would change every speed convention.

## 2 Complete bounded-profile theorem

**Theorem 1.** *Every nonconstant bounded entire profile satisfying (2) is, up to translation of  $\xi$ , exactly one of the following.*

1. If  $r_1 < r_2 < r_3$ , put

$$m = \frac{r_3 - r_2}{r_3 - r_1}, \quad k = \sqrt{\frac{r_3 - r_1}{2}}.$$

Then  $0 < m < 1$  and

$$U(\xi) = r_2 + (r_3 - r_2) \operatorname{cn}^2(k\xi; m). \quad (5)$$

2. If  $r_1 = r_2 < r_3$ , then

$$U(\xi) = r_1 + (r_3 - r_1) \operatorname{sech}^2(k\xi), \quad k = \sqrt{(r_3 - r_1)/2}. \quad (6)$$

*An upper double root, a triple root, or a cubic with only one real root has no nonconstant bounded entire profile. Constant profiles remain solutions, but do not select a traveling speed.*

**Proof.** The negative leading coefficient in (2) leaves a compact positive interval only for three simple roots, namely  $[r_2, r_3]$ , or for the lower-double-root closure  $[r_1, r_3]$ . Simple endpoints are reached and reflected in finite time; the lower double root is approached only at infinite time. Every other positive component is unbounded, proving exhaustion. For  $q = \operatorname{cn}^2(k\xi; m)$ ,

$$q_\xi^2 = 4k^2q(1 - q)(1 - m + mq).$$

Substitution of the displayed  $m$  and  $k$  is exactly (3); its derivative gives the profile ODE. The identity  $\operatorname{cn}(s; 1) = \operatorname{sech} s$  gives the homoclinic face. Thus the result is a root classification, not an ansatz selected from a numerical plot.

The baseline stops here: it fixes the all-root theorem, the plus-sign speed, and both double-root conventions before moments or executable receipts are added.

### 3 Conclusion and limitations

The result is one complete coherent-family advance: cubic-root exhaustion, cnoidal and soliton formulas, exact period observables, all declared degenerations, and the physical circle clock. It is not a nonlinear or spectral stability theorem and does not classify arbitrary KdV solutions. Its formulas are classical and re-derived source-locally; no arithmetic or literature-priority conclusion is inferred.

### References

- [1] D. J. Korteweg and G. de Vries, “On the change of form of long waves advancing in a rectangular canal, and on a new type of long stationary waves,” *Philosophical Magazine* 39 (1895), 422–443, doi:10.1080/14786449508620739.
- [2] M. Leitner and A. Mikikits-Leitner, “Nonlinear differential identities for cnoidal waves,” *Mathematische Nachrichten* 287 (2014), 2040–2056, doi:10.1002/mana.201300233.
- [3] P. D. Lax, “Integrals of nonlinear equations of evolution and solitary waves,” *Communications on Pure and Applied Mathematics* 21 (1968), 467–490, doi:10.1002/cpa.3160210503.